

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5115

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 2

Total Marks:50

Annexure A

CONCEPT MAPPING

Code of Conduct:

*I **Nissanki Sri Venkata Lakshmi Kavya(192373036)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

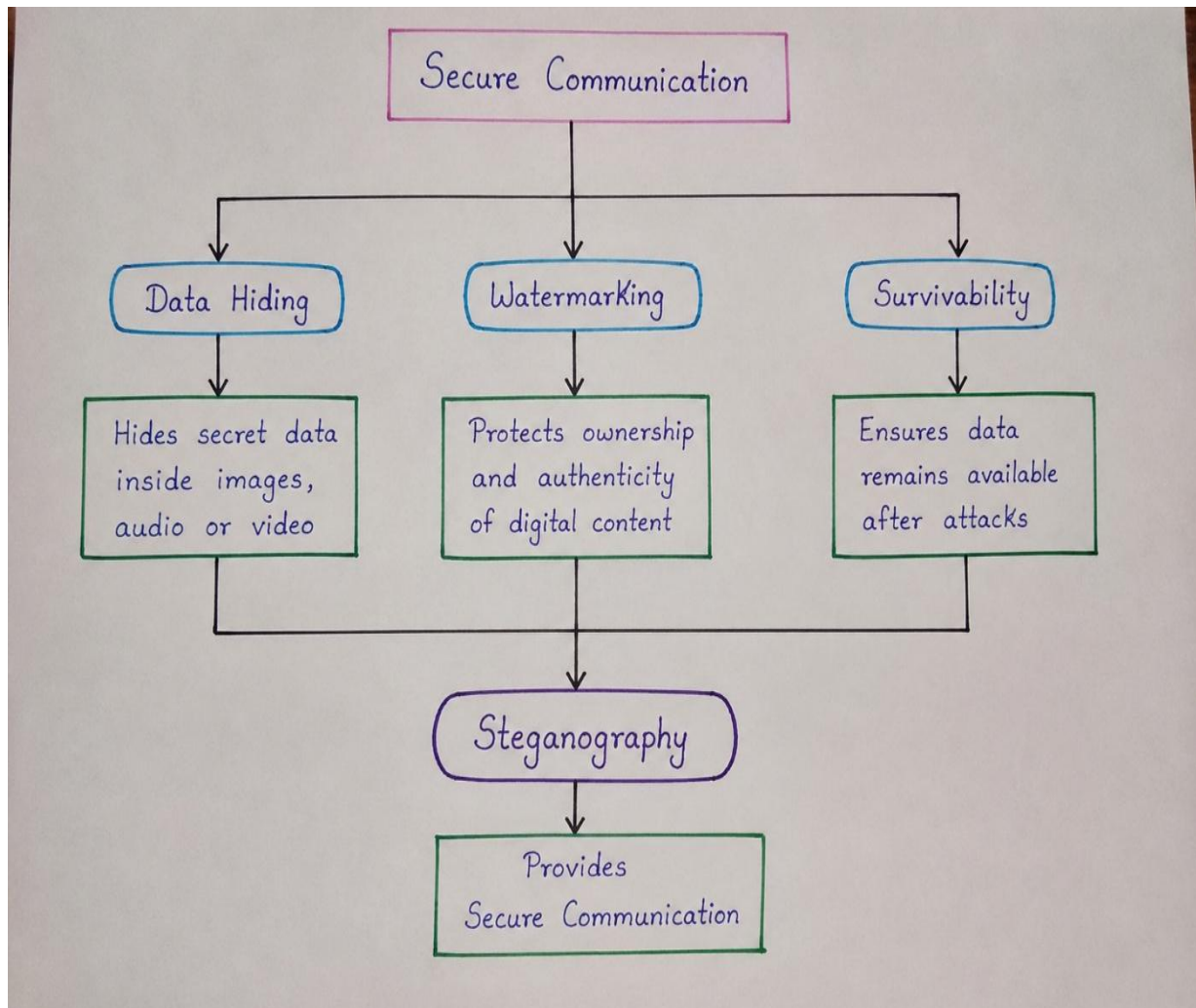
Signature of the student:

N.S.V.L.Kavya

1. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:

- Show relationships between these concepts**
- Analyze how each contributes to secure communication**
- Include one real-world application for each**

Concept Map: Steganography, Data Hiding, Watermarking, and Survivability



a. Relationship between the concepts

- **Steganography** is the main technique used to hide secret information.
- **Data Hiding** is the process of hiding confidential data inside another file such as an image, audio, or video.
- **Watermarking** is used to protect ownership and verify the originality of digital content.
- **Survivability** ensures that the hidden data remains safe and accessible even if the file is modified or attacked.
- Together, these concepts improve the security of digital communication.

b. How each contributes to secure communication

Concept	Contribution to Secure Communication
Data Hiding	Hides secret information from unauthorized users.
Watermarking	Protects ownership and verifies the authenticity of digital content.
Survivability	Ensures hidden data remains safe and can be recovered after attacks.

c. Real-World Applications

Concept	Real-World Application
Data Hiding	Hiding confidential military or government information inside digital images.
Watermarking	Adding a digital watermark to photos, videos, or movies to prevent copyright theft.
Survivability	Secure storage and transmission of medical records or military data that must remain available even after system failures or cyberattacks.

5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi

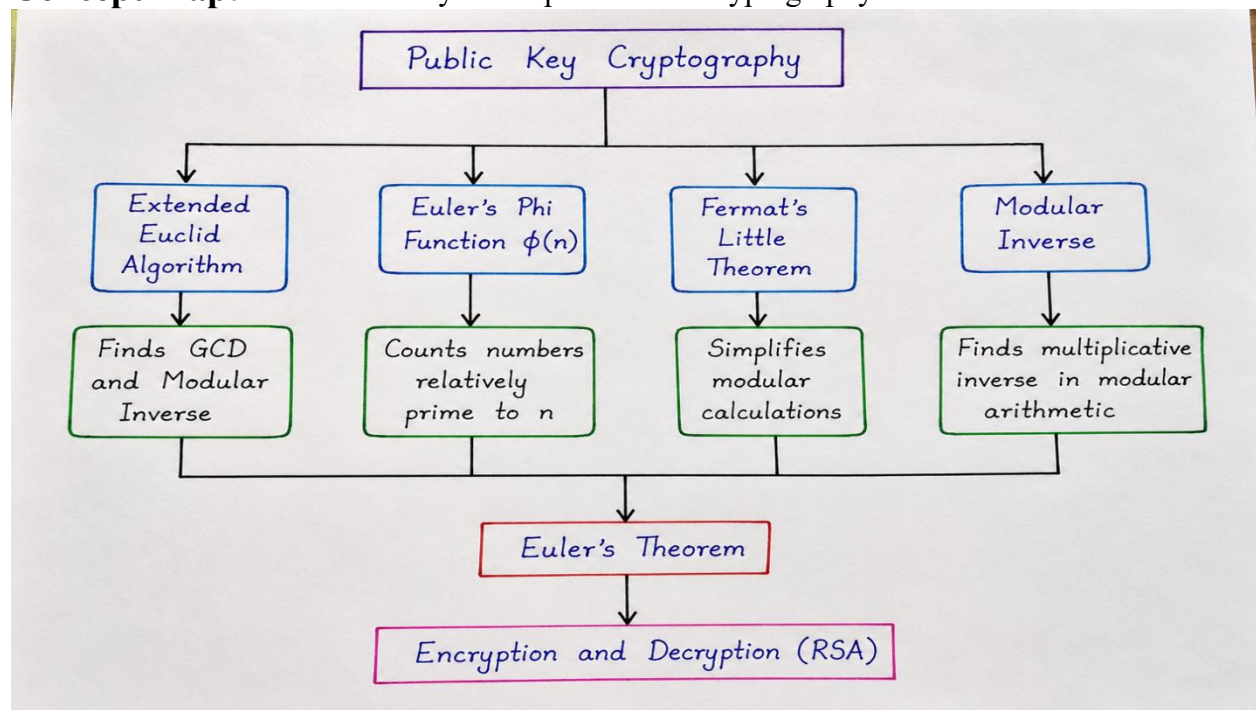
Function and Euler's Theorem. Perform the following tasks:

a. Show how each concept is mathematically connected

b. Map their role in encryption/decryption

c. Analyze which concept is most critical for public key cryptography and justify

Concept Map: Number Theory Concepts Used in Cryptography



a. How each concept is mathematically connected

- Extended Euclid Algorithm finds the GCD and is used to calculate the Modular Inverse.
- Euler's Phi Function calculates the number of integers that are relatively prime to n .
- Euler's Theorem uses the value of Euler's Phi Function.
- Fermat's Little Theorem is a special case of Euler's Theorem when the modulus is a prime number.
- All these concepts work together in RSA encryption and decryption.

b. Role in Encryption and Decryption

Concept	Role in Cryptography
Extended Euclid Algorithm	Finds the modular inverse to generate the private key.
Modular Inverse	Used to calculate the private key for decryption.
Euler's Phi Function	Used to generate public and private keys.
Euler's Theorem	Forms the mathematical basis of RSA encryption and decryption.
Fermat's Little Theorem	Simplifies modular arithmetic when the modulus is a prime number.

c. Which concept is most critical for public key cryptography? Justify.

Euler's Theorem

Reason:

Euler's Theorem is the most critical concept in public key cryptography because it forms the mathematical foundation of the RSA encryption algorithm. It is used to generate secure public and private keys and ensures that a message encrypted with the public key can be decrypted correctly only by the person who has the corresponding private key. This makes data transmission secure and prevents unauthorized users from accessing confidential information. Since RSA is one of the most widely used public key cryptography systems for secure communication, digital signatures, and online transactions, Euler's Theorem plays a vital role in maintaining the security, accuracy, and reliability of the encryption and decryption process.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understandi ng of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technol o gies	Good understandi ng with proper integration of most concepts	Basic understand in g; limited or partial integration	Poor understand in g; unable to integrate concepts
Experi ment al Design	30	Well-planned, systematic implementati on with optimal solution	Correct implemen tat ion with minor issues	Basic implementa ti on with errors or inefficiencies	Incorrect or incomplete implement ati on
Use of Tools	20	Effective and advanced use of tools/technologi es with proper configuration	Appropria te use of tools with correct execution	Limited or inconsiste nt use of tools	Incorrect or minimal use of tools
Analys is of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonabl e interpreta tio n	Basic analysis with limited insights	Poor or incorrect analysis of results
Docum entat ion	10	Clear, well structured documentation with diagrams, code, and results	Organized document ati on with necessary details	Basic documenta tio n with missing details	Poor or incomplete documenta tio n